Project Plan

Threat Modeling Tool for 6G Non-Terrestrial Networks

Advanced Project in Cyber Security (D7086E)

1. Background and Motivation

Non-Terrestrial Networks (NTN) extend network connectivity through satellites and airborne platforms. This project focuses on one transparent Low Earth Orbit (LEO) satellite access scenario based on existing 3GPP NTN studies [1].
The selected scenario connects a representative rural user device through a satellite relay and NTN gateway to a ground gNB and one operator's 5G Core. Limited ground administration is included. This provides a manageable scenario that can be studied without satellite hardware or network simulation.
The motivation is to identify and prioritise threats for a clearly documented architecture. Threats will be linked to components and interfaces plus data flows and protected assets and trust boundaries.
The project will produce a threat-modeling case study and a small web application that displays reviewed threats and explains their priority. Existing 5G NTN architecture provides the technical baseline. The report will discuss its relevance to future 6G systems.

2. Objectives

The primary objective is to develop a small web-based threat-modeling tool for the selected LEO NTN scenario. The project will:

1. Define a bounded reference architecture with documented components and interfaces plus assets and data flows and trust boundaries.
2. Identify relevant threats using STRIDE. Link each threat to the affected parts of the architecture and supporting evidence or clearly labelled assumptions.
3. Store reviewed threat records in a structured database accessible through a REST API.
4. Rank threats using documented likelihood and impact criteria. Provide a clear explanation for each score.
5. Provide a simple web interface with component and STRIDE filters plus threat details and CSV export.
6. Evaluate coverage and scoring consistency plus usability and limitations.

The core application will run on both students' computers using documented setup instructions. It will operate without an LLM or specialised hardware.
Docker packaging and an LLM query interface are optional extensions. They will be considered after the core deliverable is complete.

3. Scope

The project uses one bounded reference scenario called S1. A representative rural UE connects through one transparent LEO satellite relay and one NTN gateway to a ground gNB and one operator's 5G Core. Limited administration of the gateway and gNB is included.
This scenario is manageable for two students and can be analysed through literature and a software prototype. The study uses a static service snapshot. Mobility and delay are considered where relevant but are not simulated.

3.1 In Scope

- A reference architecture showing components and interfaces plus assets and data flows and trust boundaries.

- Threat identification using STRIDE for the selected scenario.
- A small reviewed threat database with evidence or clearly labelled assumptions.
- Ranking based on documented likelihood and impact criteria.
- A simple web application and REST API with component and STRIDE filters plus CSV export.
- Evaluation of coverage and scoring consistency plus usability and limitations.
- Documented setup allowing both students to run the core without an LLM or specialised hardware.

Optional Extensions:

After the core is complete we may add Docker packaging and further reviewed threat records within S1. An optional LLM interface may retrieve existing records through the API. It must not change records or ranking scores.

3.2 Out of Scope

- MEO and HAPS and UAVs plus inter-satellite links and regenerative payloads.
- Multiple operators and roaming.
- Full 5G Core internals and satellite command systems.
- Orbital or RF simulation plus actual satellite hardware and live attacks.
- Deployment of countermeasures in a real satellite network.
- Production deployment and formal certification.
- LLM training or fine-tuning.

4. Technical Approach

The project will combine a bounded NTN architecture with a clear threat-modeling method and a small web application. The core system will operate independently of the optional LLM interface.

4.1. Reference Architecture

The reference scenario is labelled S1. It represents rural access through one transparent LEO satellite relay connected to one operator's 5G Core interfaces. The architecture will use the existing 3GPP material listed in reference [1].

The model contains six components:

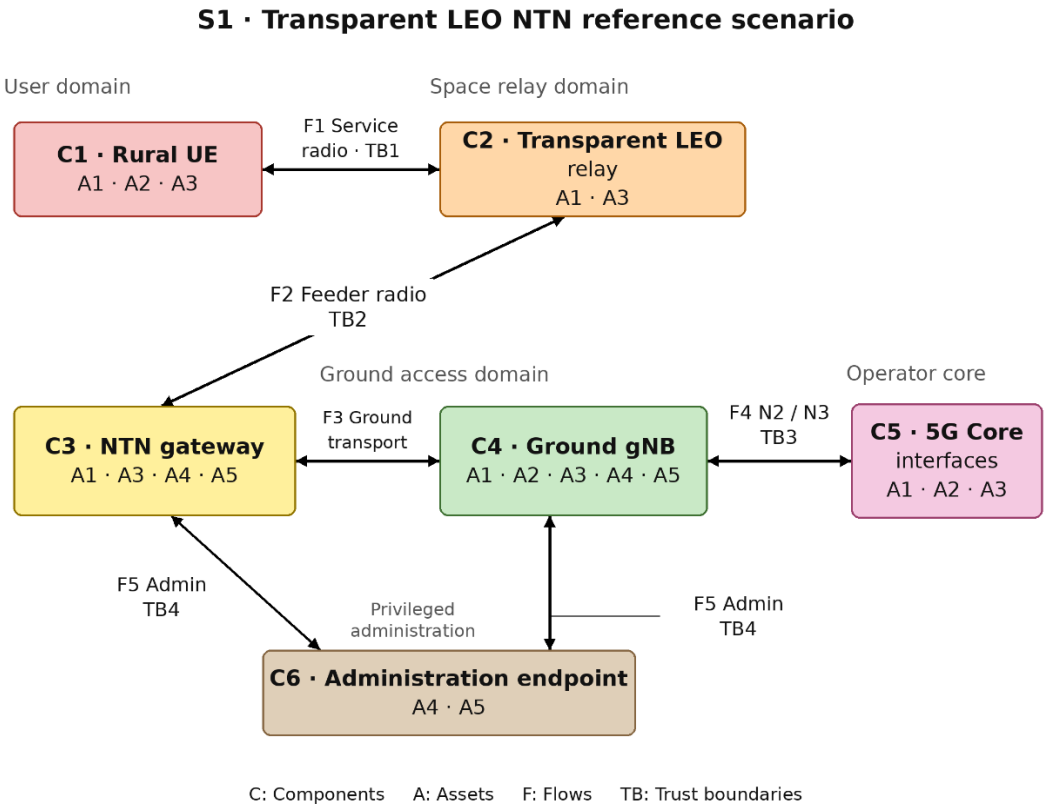
- C1 - Rural UE: The user's device.
- C2 - Transparent LEO relay: The satellite that relays radio signals.
- C3 - NTN gateway: The ground endpoint of the satellite feeder link.
- C4 - Ground gNB: The ground-based radio access network function.
- C5 - 5G Core interfaces: The connection to one operator's core network.
- C6 - Administration endpoint: The endpoint used to manage the gateway and ground gNB.

The protected assets are:

- A1: Confidentiality and integrity of user traffic.
- A2: Subscriber identity and authentication material.
- A3: Service availability and access capacity.
- A4: Ground configuration and administration credentials.
- A5: Administration change records and logs.

Initial Architecture and Data-Flow Diagram:

The diagram shows the components and flow identifiers. Asset identifiers indicate the assets considered at each component. Trust boundaries are marked on the relevant connections.



The flows and interfaces are defined as follows:

- F1 - Service radio: User traffic and signalling between C1 and C2. TB1 separates the user domain from the space relay domain.
- F2 - Feeder radio: Relayed radio signals between C2 and C3. TB2 separates the space relay domain from the ground access domain.
- F3 - Ground transport: The connection between C3 and C4 within the ground access domain.
- F4 - Core connection: N2 control signalling and N3 user traffic between C4 and C5. TB3 separates ground access from the operator core domain.
- F5 - Administration: Configuration changes and administration records between C6 and C3 or C4. TB4 marks the boundary for privileged access.

F1 and F2 are physical segments of the same access path. The gNB remains on the ground. Satellite movement and delay will be documented as assumptions without simulation.
The UE represents one user of a shared access service. Any impact on other users must be stated as an assumption in the relevant threat record. Full core internals and satellite command systems remain outside the model.

4.2. Threat Identification and Threat Database

Threat Identification

We will use the six STRIDE categories:

- Spoofing: Pretending to be another user or system.
- Tampering: Making unauthorised changes.
- Repudiation: Denying an action when reliable records are missing.
- Information Disclosure: Exposing protected information.
- Denial of Service: Preventing normal service.
- Elevation of Privilege: Obtaining permissions beyond those allowed.

The procedure will be:

1. Review the six components and five flow groups against each STRIDE category.
2. Record candidate threats with their affected assets and architecture identifiers.
3. Describe the attack mechanism and required conditions plus the possible consequence.
4. Check the description against the existing sources. Clearly label unsupported parts as assumptions.
5. Review each record together and merge duplicate threats before including it in the reviewed catalogue.

The review worksheet will contain 66 cells. Each cell will link to candidate threat records or explain why no applicable threat was identified. This provides a record of the review process without requiring 66 separate threats.

Proposed Database Structure:

Field	Purpose
Threat ID	Unique identifier such as T01
Scenario ID	Identifies the selected scenario S1
Title	Short name for the threat
Description	Attack mechanism and required conditions plus possible consequence
Architecture mappings	Affected component IDs and asset IDs plus flow IDs and trust-boundary IDs
STRIDE category	Category used to identify the threat
Likelihood	Rating from 1 to 4 with a written reason
Impact	Rating from 1 to 4 with a written reason
Risk score and band	Calculated score and its ranking band
Evidence or assumption	Existing reference number and page or section where available. Assumptions are labelled explicitly
Evidence confidence	High or medium or low with a brief explanation
Review status	Draft or reviewed with reviewer and review date

Ranking Method:

Likelihood and impact will each use a four-point scale.

Likelihood:

- 1 - Very low: Requires exceptional access or conditions with little supporting evidence.
- 2 - Low: Requires specialist equipment or privileged access plus a plausible weakness.
- 3 - Moderate: A practical attack path exists once the stated conditions are met.
- 4 - High: The attack has a low access barrier and can be repeated under the stated conditions with supporting evidence.

Impact

- 1 - Minor: Limited effect on one UE with quick recovery.
- 2 - Limited: Temporary disruption or limited information exposure.
- 3 - Major: Disruption affecting several users of the shared service or exposure of sensitive traffic or configuration.
- 4 - Severe: Sustained loss of the selected service or compromise of ground administration.

The risk score will be calculated as:

Risk score = Likelihood × Impact

The ranking bands are:

- Low: 1–3
- Medium: 4–7
- High: 8–11
- Critical: 12–16

Threats will be ordered by descending risk score. Equal scores will be ordered by descending impact and then by threat ID.

These scales are project-specific judgments. Each rating must have a written explanation. A threat will remain unscored if there is insufficient information to justify a rating. Evidence confidence will be recorded separately from the risk score.

Initial Example Threat Records:

The following records illustrate the proposed structure. Their ratings depend on the assumptions stated below. All three are drafts with low evidence confidence and await joint review.

ID	Threat	Architecture mappings	STRIDE category	Proposed score
T01	Jamming of the service radio link	C1 and C2 / A3 / F1 / TB1	Denial of Service	L2 × I3 = 6 — Medium
T02	Use of stolen administration credentials	C3 and C4 and C6 / A3 and A4 and A5 / F5 / TB4	Spoofing	L2 × I4 = 8 — High
T03	Passive observation of radio traffic	C1 and C2 / A1 / F1 / TB1	Information Disclosure	L2 × I2 = 4 — Medium

T01 - Jamming of the Service Radio Link

- Required conditions: An attacker can access the relevant radio environment and has suitable interference equipment.
- Likelihood reason: Specialist equipment and a suitable location are required.
- Impact reason: The draft assumes interference could disrupt access for several users sharing the affected service.
- Evidence status: Assumption pending review against the existing sources.

T02 - Use of Stolen Administration Credentials

- Required conditions: An attacker obtains valid credentials and can reach the administration interface.
- Likelihood reason: Both credential compromise and interface access are required.
- Impact reason: The draft assumes the compromised account can change important ground configuration and interrupt the selected service.
- Evidence status: Assumption pending review against the existing sources.

T03 - Passive Observation of Radio Traffic

- Required conditions: An attacker has suitable receiving equipment and access to the relevant signal.
- Likelihood reason: Observation requires specialist equipment and useful exposed information.
- Impact reason: The draft assumes visible metadata could reveal communication patterns. It does not assume that encrypted user traffic can be decrypted.
- Evidence status: Assumption pending review against the existing sources.

4.3. Web Application and API

The core prototype will consist of a browser interface connected to an API and a local database. The application will allow users to:

- View the selected S1 architecture.
- Browse reviewed threat records.
- Filter records by component and STRIDE category.
- View threat details with ranking reasons and evidence or assumption labels.
- Export the selected records as CSV.

The API and browser interface will use the same filtering and ranking rules. The implementation will check that architecture mappings refer to valid identifiers and that threat IDs are unique. Both students must be able to run the core application using documented setup instructions without an LLM or specialised hardware. Docker packaging is an optional extension.

4.4. Optional LLM Interface

An LLM interface may be added after the core application and evaluation are complete. The interface will retrieve existing threat records through the API. It will provide a read-only way to explore the reviewed catalogue. It must not create or modify threat records or change ranking scores.

All core functions will remain available without Marius's private hardware or LLM environment. LLM training and fine-tuning are outside the project scope.

5. Work Packages and Timeline

The work plan prioritises the core deliverable and preparation for the mandatory mid-seminar on 21 October 2026. Docker packaging and the LLM interface remain optional.
All dates below are in 2026. Meeting times use Stockholm time. Dates after the mid-seminar are internal targets and will be adjusted to the confirmed course deadlines.

5.1. Division of Responsibilities

- John: Lead the review of existing literature and architecture analysis plus threat identification and browser-interface development.
- Marius: Lead the threat-database design and API development plus ranking implementation and CSV export. Provide the exact details of the sources already selected.
- Both students: Review threat records and scoring decisions. Complete evaluation and documentation together. Prepare the report and presentations jointly.

Each student will review the other student's work and remain able to explain the complete core system.

5.2. Dated Work Plan

The supervisor requested the first revised project plan and team charter by 5 October at 12:00 noon. Any outstanding documents will be jointly reviewed and submitted before the meeting on 9 October.

Dates	Planned work	Responsibility	Expected deliverable
7–8 October	Revise the plan and charter. Prepare the bounded scenario and architecture diagram plus the database structure and three draft threat records	John leads the architecture. Marius leads the database. Both review the documents	Revised documents and initial progress presentation with specific details for the existing references
9 October — proposed meeting at 10:00–10:30	Present progress and discuss the scenario and method plus responsibilities and evaluation. Confirm final course deadlines	Both students	Recorded supervisor feedback and an updated work plan
10–14 October	Develop the STRIDE review worksheet and initial threat catalogue. Implement the database and basic API plus ranking rules	John leads threat identification. Marius leads implementation. Both review records	Initial reviewed catalogue with score explanations and evidence or assumption labels
15–20 October	Connect the browser interface to the API. Add component and STRIDE filters plus CSV export. Check ranking and setup on both computers	John leads the interface. Marius leads integration. Both check the core	Working core prototype and preliminary evaluation results. Presentation rehearsed by 20 October
21 October	Present the project at the mandatory mid-seminar	Both students	Mid-seminar presentation and demonstration with documented progress and remaining work

22–28 October	Address seminar feedback. Complete the coverage and scoring reviews. Conduct a small usability review and fix blocking issues	Both students	Evaluation results and an improved core application with documented limitations
29 October–4 November	Complete the report draft and setup instructions. Review the code and database. Prepare the final demonstration	Both students	Final report draft and reproducible core package ready for review
Confirmed course deadlines	Submit the final deliverables and give the final presentation	Both students	Final report and application plus presentation

5.3. Mid-Seminar Deliverables

By 21 October we aim to have:

- A clearly bounded S1 scenario and architecture diagram showing components and interfaces plus assets and trust boundaries.
- A documented STRIDE procedure and likelihood–impact ranking method.
- An initial reviewed threat catalogue with traceable mappings and explained scores.
- A working browser interface and API with the agreed filters and ranking plus CSV export.
- Setup instructions checked on both students' computers without an LLM or specialised hardware.
- Preliminary coverage and scoring-consistency results plus a clear account of limitations and remaining work.

Optional extensions will only be started after the core application and evaluation are complete and sufficient time remains.

6. Main Risks and Mitigations

Risks will be reviewed during weekly team meetings and supervisor check-ins. The main risks and proposed responses are:

- **Scope growth:** Additional scenarios or features could delay the core deliverable. We will keep the work within the selected S1 scenario. Material scope changes will be discussed with the supervisor.
- **Limited threat evidence**: Some threat descriptions may lack clear support in the existing sources. We will verify the source details and record supporting pages or sections where available. Assumptions will be clearly labelled. Evidence gaps will be documented as limitations.
- **Inconsistent ranking:** John and Marius may interpret likelihood and impact differently. We will use the same rating criteria and write a reason for each score. Differences will be discussed during joint review and the agreed reasoning will be recorded.
- **Implementation delays:** Problems connecting the database and API to the browser interface could reduce time for evaluation. We will build and connect the basic core early. Optional Docker packaging and the LLM interface will be postponed until the core and evaluation are complete.
- **Setup problems:** The application may work on one student's computer but fail on the other. We will document the required software and setup steps. Both students will check the core without an LLM or specialised hardware before the mid-seminar.

- **Limited time or availability:** Delayed tasks could affect the mandatory mid-seminar on 21 October. We will track progress weekly and raise blockers early. Tasks may be reassigned by agreement. The remaining course deadlines will be confirmed with the supervisor and added to the work plan.

7. Expected Outcomes

The project is expected to produce the following core deliverables:

- Documented reference architecture: A clearly bounded S1 scenario showing the selected components and interfaces plus assets and data flows and trust boundaries. Architecture assumptions and exclusions will be stated.

- Defined threat-modeling method: A documented STRIDE procedure and review worksheet with likelihood–impact ranking criteria and written score explanations.

- Reviewed threat database: A small catalogue of threats linked to the architecture. Each reviewed record will include supporting evidence or clearly labelled assumptions plus its review status.

- Working web application and API: A core prototype that displays the scenario and reviewed threats. It will support component and STRIDE filters plus ranked results and CSV export.

- Evaluation results: An assessment of coverage and scoring consistency plus usability and limitations. Evidence gaps and unresolved issues will be reported.

- Reproducible project package: Code and database files with setup instructions that allow both students to run the core without an LLM or specialised hardware.

- Academic deliverables: A mid-seminar presentation and demonstration on 21 October 2026 followed by the final report and presentation at the confirmed course deadlines.

Docker packaging and the LLM interface may be added if the core application and evaluation are complete and time remains.

8. Evaluation Approach

Evaluation will assess coverage and scoring consistency plus functional correctness and usability. It will use the selected S1 scenario and the reviewed threat catalogue.
Preliminary results will be presented at the mid-seminar on 21 October 2026. The full evaluation is planned for 22–28 October 2026.

8.1. Coverage

We will review the six components and five flow groups against the six STRIDE categories using the 66-cell worksheet.

Each cell must link to a candidate threat record or explain why no applicable threat was identified. We will also check that every asset and trust boundary has been considered.

Each reviewed threat record will be checked for valid architecture mappings and complete required fields. Supporting evidence or assumptions must be clearly identified.

Completion criterion: All worksheet cells have a recorded outcome. Every asset and trust boundary has been reviewed. Evidence gaps are documented.

This measures the coverage of our review within S1. It does not establish that every possible NTN threat has been identified.

8.2. Scoring Consistency

Both students will independently score up to five shared threat records using the likelihood and impact criteria in Section 4.2.

We will record the initial ratings and count agreements for likelihood and impact separately. Differences will be discussed before agreeing on final ratings and written reasons.
Completion criterion: Every reviewed record has justified likelihood and impact ratings. Initial scoring differences and the reasons for the final decisions are documented.

8.3. Functional Correctness

We will check that:

- Component and STRIDE filters return the expected records.
- Risk scores equal likelihood multiplied by impact.
- Ranking bands and tie-breaking rules follow Section 4.2.
- Unscored records are clearly identified.
- The browser interface and API return matching results for the same filters.
- CSV exports contain the selected records and their stored values.

Expected results will be checked against the database and documented ranking rules.

Completion criterion: The agreed core functions produce the expected results. Any failed checks are corrected and checked again.

8.4. Usability

One person outside the project team will attempt four tasks without step-by-step assistance:

1. View the S1 scenario.
2. Filter threats by a selected component and STRIDE category.
3. Open a threat record and find its score explanation and supporting evidence or assumption.
4. Export the selected records as CSV.

We will record task completion and errors plus any help required. Feedback will guide small improvements to the interface and instructions.

Success criterion: All four tasks can be completed without a blocking problem. Any unsuccessful tasks and resulting changes are documented.

8.5. Reproducibility

Both students will follow the setup instructions on their own computers and run the core application without an LLM or specialised hardware.

They will check that the scenario and threat catalogue are available and that filtering and ranking plus CSV export operate correctly.

Completion criterion: Both students can install and run the core using the documented steps.

8.6. Limitations

The final report will discuss:

- The restriction to one representative S1 scenario.
- Gaps in the existing sources and reliance on labelled assumptions.
- The judgment involved in likelihood and impact ratings.
- The limited usability review with one participant.
- The absence of live attack testing and real satellite or radio validation.

The report will state which evaluation criteria were met and which issues remain unresolved. Findings will be limited to the selected scenario and prototype.

References

[1] 3GPP, “Study on New Radio (NR) to support non terrestrial networks,” TR 38.811 / TR 38.821, and subsequent security related work items.

[2] ETSI/3GPP material on NTN architecture and the integration of LEO/MEO, HAPS and UAV platforms into 5G Advanced and 6G systems.

[3] Surveys and conference papers analysing jamming, spoofing, eavesdropping and DoS threats in satellite and aerial networks (e.g. recent IEEE and arXiv works on 6G NTN security).

[4] Studies on inter satellite link security and the consequences of regenerative payloads that move base station functions on board.

[5] Tutorial and overview papers on multi layer NTN architectures toward 6G (space air ground integration).

[6] Work on physical layer security, zero trust ideas and AI assisted detection specifically targeted at NTN environments.

Note: The complete bibliographic references will be provided in the final report after deciding on the papers to use in the database of threats.